

Spearman Rank Correlation

In a study of the relationship between two variables, the use of measures of **correlation** assumes that neither is functionally dependent upon the other. So, for example, we might ask whether body weight is related to height in 35-year-old men; or whether examination scores in music theory are related to examination scores in mathematics for first-year college students; or whether the blood levels of two steroid hormones are related in 20-year-old women. A quantitative measure of the strength of the correlation is a correlation coefficient, which expresses how closely a change in the magnitude of one of the variables is accompanied by a change in the magnitude of the other variable. This is also referred to as a measure of association or of correspondence (*see Association, Measures of*).

If the distributions underlying the two variables are far from **bivariate normal**, or if the data are ordinal (e.g. we know relative magnitudes—such as man A is taller than man C but shorter than man D—but we do not know their actual heights) (*see Ordered Categorical Data*), then **nonparametric** correlation techniques should be employed to test hypotheses about the relationship between variables or to set **confidence limits** around the correlation coefficients. Nonparametric correlation also is less sensitive to **outliers** than is its parametric analog. The underlying assumptions for nonparametric correlation are that the n pairs of ratio, interval, or ordinal data (*see Measurement Scale*) constitute a **random sample** and that the two members of each of the n pairs of data are measurements taken on the same subject.

Among the correlation coefficients proposed by Charles Spearman [19, 20] is a commonly used nonparametric correlation measure that Maurice Kendall formally associated with Spearman's name a quarter of a century later [14], and that is one of the oldest statistics based on ranks. The Spearman rank coefficient computed for a sample of data is typically designated as r_s .

If each of the n measurements of one of the variables is denoted as X_i (i.e. $X_1, X_2, \dots, X_n$), then $\mathcal{R}(X_i)$ may represent the **rank** of X_i , where each rank is an integer, from 1 through n , indicating relative magnitude. The measurements may be ranked from

Table 1

Person, i	1	2	3	4	5
Height (m), Y_i	1.59	1.66	1.82	1.73	1.91
Weight (kg), X_i	75.8	77.2	89.3	72.2	81.5
Rank of height, $\mathcal{R}(X_i)$	1	2	4	3	5
Rank of weight, $\mathcal{R}(Y_i)$	2	3	5	1	4
$d_i = \mathcal{R}(X_i) - \mathcal{R}(Y_i)$	-1	-1	-1	2	1
d_i^2	1	1	1	4	1

high to low (e.g. rank 1 indicates the tallest person, rank 2 the next tallest, and so on, with rank n the shortest) or from low to high (rank 1 denotes the shortest and rank n the tallest). Similarly, each of the n measurements of the second variable may be denoted as Y_i (i.e. $Y_1, Y_2, \dots, Y_n$), and $\mathcal{R}(Y_i)$ would denote the rank of Y_i , where the sequence of ranking (either high to low or low to high) is the same as for $\mathcal{R}(X_i)$. This is shown in Table 1.

An $r_s = 0$ ("no correlation") indicates that the magnitudes of the ranks of one variable are independent of the magnitudes of the ranks of the second variable. A positive value of r_s ("positive correlation") indicates that the $\mathcal{R}(X_i)$ s tend to increase as the $\mathcal{R}(Y_i)$ s increase; a negative r_s ("negative correlation") indicates that the $\mathcal{R}(X_i)$ s tend to decrease as the $\mathcal{R}(Y_i)$ s increase.

If the sequence of ranks were identical for the two variables, we would say that there was a perfect positive correlation, and $r_s = 1.0$. This would occur, for example, if five pairs of data had these ranks:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{array}$$
 or these:

$$\begin{array}{ccccc} 1 & 2 & 4 & 3 & 5 \\ 1 & 2 & 4 & 3 & 5 \end{array}$$

A perfect negative correlation (where $r_s = -1.0$) would be one in which the magnitudes of the ranks for one variable vary inversely with the sizes of the ranks of the second; for example,

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 2 & 1 \end{array}$$

Computing the Coefficient

The widely used parametric correlation coefficient, known as the Pearson product-moment correlation

2 Spearman Rank Correlation

coefficient (*see Correlation*), is defined as

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\left[\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2 \right]^{1/2}}, \quad (1)$$

and commonly computed as

$$r = \left(\sum XY - \frac{\sum X \sum Y}{n} \right) / \left\{ \left[\sum X^2 - \frac{(\sum X)^2}{n} \right] \left[\sum Y^2 - \frac{(\sum Y)^2}{n} \right] \right\}^{1/2}, \quad (2)$$

where $\bar{X}$ and $\bar{Y}$ are the means of the X_i s and the Y_i s, respectively, and where the summations ($\sum$) are each over all n data.

The Spearman **rank correlation** coefficient, r_s , may be obtained by subjecting the ranks, instead of the raw measurements, to the above calculations. For the example above, and substituting $\mathcal{R}(X_i)$ for X_i and $\mathcal{R}(Y_i)$ for Y_i :

$$\sum X_i = 1 + 2 + 4 + 3 + 5 = 15,$$

$$\sum (X_i)^2 = 1^2 + 2^2 + 4^2 + 3^2 + 5^2 = 55,$$

$$\sum Y_i = 2 + 3 + 5 + 1 + 4 = 15,$$

$$\sum (Y_i)^2 = 2^2 + 3^2 + 5^2 + 1^2 + 4^2 = 55,$$

and

$$\begin{aligned} \sum (X_i Y_i) &= (1)(2) + (2)(3) + (4)(5) + (3)(1) \\ &\quad + (5)(4) = 51. \end{aligned}$$

Then,

$$\bar{X} = 15/5 = 3 \quad \text{and} \quad \bar{Y} = 15/5 = 3,$$

and

$$\begin{aligned} \sum (X_i - \bar{X})^2 &= (1-3)^2 + (2-3)^2 + (4-3)^2 \\ &\quad + (3-3)^2 + (5-3)^2 = 10, \\ \sum (Y_i - \bar{Y})^2 &= (2-3)^2 + (1-3)^2 + (5-3)^2 \\ &\quad + (3-3)^2 + (4-3)^2 = 10, \end{aligned}$$

$$\begin{aligned} \sum (X_i - \bar{X})(Y_i - \bar{Y}) &= (1-3)(2-3) + (2-3) \\ &\quad \times (3-3) + (4-3)(5-3) \\ &\quad + (3-3)(1-3) + (5-3) \\ &\quad \times (4-3) = 6. \end{aligned}$$

Eq. (1) yields

$$r_s = \frac{6}{[(10)(10)]^{1/2}} = \frac{6}{10} = 0.60,$$

while (2) yields

$$\begin{aligned} r_s &= \frac{51 - (15 \times 15)/5}{[(55 - (15)^2/5)(55 - (15)^2/5)]^{1/2}} \\ &= \frac{6}{[(10)(10)]^{1/2}} = 0.60. \end{aligned}$$

As the sum of integers 1 through n (i.e. the sum of all n ranks) is $n(n+1)/2$, (2) employed for Spearman rank correlation may be written as

$$r_s = \frac{\sum \mathcal{R}(X_i)\mathcal{R}(Y_i) - n(n+1)^2/4}{\left\{ \left[\sum \mathcal{R}(X_i)^2 - n(n+1)^2/4 \right] \times \left[\sum \mathcal{R}(Y_i)^2 - n(n+1)^2/4 \right] \right\}^{1/2}}. \quad (3)$$

Also, as the sum of the squares of all n ranks is $n(n+1)(2n+1)/6$, (2) using ranks can be reduced to

$$r_s = \frac{12 \left[\sum \mathcal{R}(X_i)\mathcal{R}(Y_i) - n(n+1)^2/4 \right]}{n^3 - n} \quad (4)$$

or

$$r_s = \frac{12 \sum \mathcal{R}(X_i)\mathcal{R}(Y_i)}{n^3 - n} - \frac{3(n+1)}{n-1}. \quad (5)$$

Alternatively, the difference, d_i , for each pair of ranks may be obtained, and the following equation used:

$$r_s = 1 - \frac{6 \sum d_i^2}{n^3 - n}, \quad (6)$$

which, for the above example, is

$$\begin{aligned} r_s &= 1 - \frac{6(1+1+1+4+1)}{5^3 - 5} \\ &= 1 - \frac{48}{120} = 1 - 0.40 = 0.60. \end{aligned}$$

Eq. (6) is most commonly encountered in textbooks, but (1) is very convenient on a computer.

Instead of the differences between pairs of ranks, one may use the sums of the ranks for each pair [15, p. 227; [20]]:

$$r_s = \frac{6 \sum S_i^2}{n^3 - n} - \frac{7n + 5}{n - 1}, \quad (7)$$

where

$$S_i = \mathcal{R}(Y_i) + \mathcal{R}(X_i). \quad (8)$$

It can be shown [13] that in bivariate normal populations the Pearson correlation coefficient, ρ , is

$$\rho = 2 \sin \left(\frac{\pi}{6} \rho_s \right). \quad (9)$$

Tied Ranks

If two or more data have the same value, then they are said to be “tied”, and each of their ranks may be set equal to the mean of the ranks of the positions they occupy in the ordered data set. For example, in the data set 70, 74, 74, 78, and 79 kg, data 2 and 3 are tied; the mean of 2 and 3 is 2.5, so the ranks of the five data are 1, 2.5, 2.5, 4, and 5. In the data set 1.6, 1.7, 1.9, 1.9, and 1.9 m, data 3, 4, and 5 are tied; the mean of 3, 4, and 5 is 4, so the ranks of the five data are 1, 2, 4, 4, 4.

Then these ranks would be subjected to (1), or, equivalently, the following calculation [9, p. 366] would be used as an alternative to (4):

$$r_s = \left(12 \left[\sum \mathcal{R}(X_i) \mathcal{R}(Y_i) - n(n+1)^2/4 \right] \right) / \left\{ \left[(n^3 - n) - 12 \sum t_x \right] \times \left[(n^3 - n) - 12 \sum t_y \right] \right\}^{1/2} \quad (10)$$

and the following [12, p. 38; [20]] is an alternative to (6):

$$r_s = \left((n^3 - n)/6 - \sum d_i^2 - \sum t_x - \sum t_y \right) / \left\{ \left[(n^3 - n)/6 - 2 \sum t_x \right] \times \left[(n^3 - n)/6 - 2 \sum t_y \right] \right\}^{1/2}, \quad (11)$$

where

$$\sum t_x = \frac{\sum (t_i^3 - t_i)}{12}, \quad (12)$$

where t_i is the number of tied values of X in a group of ties (two in the paragraph above) and the summation is over all groups of tied X s, and

$$\sum t_y = \frac{\sum (t_i^3 - t_i)}{12}, \quad (13)$$

where t_i is the number of tied values of Y in a group of ties (three in the paragraph above) and the summation is over all groups of tied Y s. Similarly, the following [21] is an alternative to (7):

$$r_s = \frac{\left\{ \sum S_i^2 - [(n^3 - n)/6][(7n + 5)/(n - 1)] - \sum t_x - \sum t_y \right\}}{\left\{ \left[\frac{(n^3 - n)/6 - 2 \sum t_x}{\times [(n^3 - n)/6 - 2 \sum t_y]} \right]^{1/2} \right\}}, \quad (14)$$

If $\sum t_x$ and $\sum t_y$ are both zero, then (10) is equivalent to (4), (11) equals (6), and (14) is equivalent to (7). The results from these equations for tied and nontied data are noticeably different only if there are many ties.

Testing Hypotheses

The r_s calculated from a sample of data is an estimate of ρ_s , the Spearman rank correlation coefficient that would be obtained from the entire population of data from which that sample came; ρ_s is sometimes called “Spearman’s rho”.

A common desire in rank correlation analysis is to test the **null hypothesis** that there is no correlation in the population between the paired ranks, i.e. we wish to test the two-tailed hypotheses $H_0 : \rho_s = 0$ vs. $H_a : \rho_s \neq 0$ (see **Hypothesis Testing**). There are many tables of critical values of r_s , and if r_s is greater than the relevant critical value, then H_0 is rejected. The use of $\sum d_i^2$, instead of r_s , as the test statistic for rank-correlation testing is sometimes called the “Hotelling–Pabst test” [10]. $\sum d_i^2$ is small when r_s is large, and H_0 is rejected if $\sum d_i^2$ is *less than* the critical value. Published tables offer critical values for various sample sizes, n , and levels of significance,

4 Spearman Rank Correlation

α . The most extensive of such tables for r_S are those of Zar [22, Appendix, pp. 115–116] and, with slight improvements, of Ramsey [17]. If there are tied data, critical values are only approximate. It should be noted that computer software packages may use approximations that are not as accurate as published tables.

One-tailed hypotheses may also be considered. For $H_0 : \rho_S \leq 0$ vs. $H_a : \rho_S > 0$, H_0 is rejected if r_S is positive and greater than the critical value for $\alpha/2$. For $H_0 : \rho_S \geq 0$ vs. $H_a : \rho_S < 0$, H_0 is rejected if r_S is negative and its absolute value is greater than the critical value for $\alpha/2$. If n is larger than that in these large tables, then one may compute

$$t = \frac{r_S}{s}, \quad (15)$$

where s , the standard error of r_S , is

$$s = \left(\frac{1 - r^2}{n - 2} \right)^{1/2}, \quad (16)$$

for which two- and one-tailed critical values of t (**Student's t distribution**), for $df = n - 2$, are readily found. Equivalently, one may employ

$$F = \frac{1 + |r_S|}{1 - |r_S|} \quad (17)$$

[2], referring to two- or one-tailed critical values of the **F distribution** for numerator and denominator $df = n - 2$. Using t or F is valid even with tied data, and is preferable in any case to employing the normal approximation,

$$Z = r_S(n - 1)^{1/2}. \quad (18)$$

The Fisher Transformation

If n is at least moderately large, the Spearman correlation coefficient may be subjected to the Fisher z transformation by

$$z = 0.5 \ln \frac{1 + r_S}{1 - r_S}, \quad (19)$$

and there are tables, e.g. [22, Appendix pp. 110–111], available to obviate the need to perform this computation. With this transformed value, one may test null hypotheses that ρ_S equals some value other than zero;

i.e. $H_0 : \rho = \rho_0$ vs. $H_a : \rho \neq \rho_0$, where $\rho_0 \neq 0$. This is done via

$$Z = \frac{z - \zeta_0}{\sigma_z}, \quad (20)$$

where z is the transform of r_S ; ζ_0 is the transform of the hypothesized coefficient, ρ_0 ; the standard error of z is approximated by

$$\sigma_z = \left(\frac{1.060}{n - 3} \right)^{1/2} \quad (21)$$

[7, 8], and Z is a normal deviate. In this fashion both two-tailed and one-tailed hypotheses may be tested.

Confidence Limits

The z transformation also allows the setting of approximate $1 - \alpha$ confidence limits for ρ_S . The confidence limits for the z transformation are

$$z \pm Z_\alpha \sigma_z, \quad (22)$$

where $Z_\alpha = t_\alpha(\infty)$. Then, the lower confidence limit of the transformation, $L_1 = z - Z_\alpha \sigma_z$, is converted to the lower confidence limit of ρ_S by

$$\frac{\exp(2L_1) - 1}{\exp(2L_1) + 1}, \quad (23)$$

and the upper confidence limit of the transformation, $L_2 = z + Z_\alpha \sigma_z$, is converted to the upper confidence limit of ρ_S by substituting L_2 for L_1 in (23) above. Published tables, e.g. [22, Appendix, pp. 112–114], execute these conversions.

Power of Testing

For data that meet the normality assumptions of parametric correlation analysis, use of the Spearman method has a relative **efficiency** of $9/\pi^2 = 0.912$ compared with the parametric procedure for testing hypotheses about the population correlation coefficient [10]. For other data distributions, the Spearman procedure may perform even better. The power of hypothesis tests for ρ_S , and the determination of the minimum sample size needed to achieve a desired power, may be approximated by an adaptation of the procedures of Cohen [4, p. 546], as shown by Zar [22, pp. 379–380, 392].

Other Rank Correlation Measures

The Kendall rank correlation coefficient [11, 12] is the other commonly encountered rank correlation measure (see **Rank Correlation**). It is often referred to as Kendall's tau, with the population parameter designated as τ and the sample estimate of τ denoted as $\hat{\tau}$, t , T , or (unfortunately) τ . Whereas $\hat{\tau}$ is an unbiased estimate of τ , r_S is a biased estimate of ρ_S , with $E(r_S) = [3\tau + (n-2)\rho_S]/(n+1)$ [6], but this bias disappears rapidly as n increases.

The two rank-correlation procedures have different underlying premises and influences (e.g. r_S is more affected by larger d_i s), so they do not necessarily yield identical coefficients, $\hat{\tau}$ and r_S ; indeed, data sets may have the same $\hat{\tau}$ s yet different r_S s. However, there is a very strong correlation between the two coefficients, and they each may range between -1.0 and 1.0 . Daniels [5] found the relationship

$$-(n-2) \leq 3n\hat{\tau} - 2(n+1)r_S \leq (n-2),$$

which, for large n , is

$$-1 \leq 3\hat{\tau} - 2r_S \leq 1.$$

A better relationship was proved by Durbin & Stuart [6] to be

$$\frac{3n\hat{\tau} - (n-2)}{2(n+1)} \leq r_S \leq 1 - \frac{(1-\hat{\tau})}{2(n+1)} \times [(n-1)(1-\hat{\tau}) + 4].$$

Whether r_S or $\hat{\tau}$ is preferable depends upon the criteria employed to make the judgment; Chow et al. [3] judged r_S to be the preferable estimator.

Spearman's [19, 20] introduction of correlation between ranks was accompanied by a correlation measure of which he was fond, the "Spearman footrule", based upon $\sum |\mathcal{R}(X_i) - \mathcal{R}(Y_i)|$ instead of $\sum [\mathcal{R}(X_i) - \mathcal{R}(Y_i)]^2$. This measure is less useful than r_S in statistical analysis and is no longer encountered.

If one's interest is predominantly in the correlation among the largest (or smallest) members in the two populations, then the weighted rank correlation concept [18, 16; see also [22], pp. 392–395] might usefully be employed.

The Spearman rank correlation coefficient, r_S , is related to the Kendall coefficient of concordance, W , when there are two sets of ranks, as

$$W = \frac{(r_S + 1)}{2}. \quad (24)$$

Basler [1] discusses a relationship between $\sum d_i^2$ and the **chi-square test** statistic in a fourfold **contingency table** with ordinal marginal categories (see **Two-by-Two Table**).

References

- [1] Basler, H. (1988). Equivalence between tie-corrected Spearman test and a chi-square test in a fourfold contingency table, *Metrika* **35**, 203–209.
- [2] Cacoullos, T. (1965). A relation between the t and F distributions, *Journal of the American Statistical Association* **60**, 528–531.
- [3] Chow, B., Miller, J.E. & Dickinson, P.E. (1974). Extensions of Monte Carlo comparison of some properties of two rank correlation coefficients in a small sample, *Journal of Statistical Computation and Simulation* **3**, 189–195.
- [4] Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences*, 2nd Ed. Lawrence Earlbaum, Hillsdale.
- [5] Daniels, H.E. (1950). Rank correlation and population models, *Journal of the Royal Statistical Society, Series B* **12**, 171–181.
- [6] Durbin, J. & Stuart, A. (1951). Inversions and rank correlation coefficients, *Journal of the Royal Statistical Society, Series B* **13**, 303–309.
- [7] Fieller, E.C., Hartly, H.O. & Pearson, E.S. (1957). Tests for rank correlation coefficients, *Biometrika* **44**, 470–481.
- [8] Fieller, E.C., Hartly, H.O. & Pearson, E.S. (1961). Tests for rank correlation coefficients. II, *Biometrika* **48**, 29–40.
- [9] Gibbons, J.D. & Chakraborti, S. (1992). *Nonparametric Statistical Inference*, 3rd Ed. Marcel Dekker, New York.
- [10] Hotelling, H. & Pabst, M.R. (1936). Rank correlation and tests of significance involving no assumption of normality, *Annals of Mathematical Statistics* **7**, 29–43.
- [11] Kendall, M.G. (1938). A new measure of rank correlation, *Biometrika* **30**, 81–93.
- [12] Kendall, M.G. (1962). *Rank Correlation Methods*, 3rd Ed. Charles Griffin, London.
- [13] Kruskal, W.H. (1958). Ordinal measures of association, *Journal of the American Statistical Association* **53**, 814–861.
- [14] Lovie, A.D. (1995). Who discovered Spearman's rank correlation?, *British Journal of Mathematical and Statistical Psychology* **48**, 255–269.
- [15] Meddis, R. (1984). *Statistics Using Ranks: A Unified Approach*. Basil Blackwell, Oxford.
- [16] Quade, D. & Salama, I. (1992). A survey of weighted rank correlation, in *Order Statistics and Nonparametrics: Theory and Applications*, P.K. Sen & I. Salama, eds. Elsevier, New York, pp. 213–224.
- [17] Ramsey, P.H. (1988). Critical values for Spearman's rank order correlation, *Journal of Educational Statistics* **14**, 245–253.

6 Spearman Rank Correlation

- [18] Salama, I. & Quade, D. (1982). A nonparametric comparison of two multiple regressions by means of a weighted measure of correlation, *Communications in Statistics – Theory and Methods* **11**, 1185–1195.
- [19] Spearman, C. (1904). The proof and measurement of correlation between two things, *American Journal of Psychology* **15**, 72–101.
- [20] Spearman, C. (1906). “Footrule” for measuring correlation, *British Journal of Psychology* **2**, 89–108.
- [21] Thomas, G.E. (1989). A note on correcting for ties with Spearman’s ρ , *Journal of Statistical Computation and Simulation* **31**, 37–40.
- [22] Zar, J.H. (1996). *Biostatistical Analysis*, 3rd Ed. Prentice-Hall, Upper Saddle River.

JERROLD H. ZAR